ANNEX I. Kyrgyz Republic: Debt Sustainability Analysis Update

The Kyrgyz Republic[1] remains at a moderate risk of debt distress. Stress tests suggest that the country is still vulnerable to large exogenous shocks. Fiscal consolidation and prudent use of concessional loans, in particular for large-scale public investment programs, are key to keeping the debt outlook sustainable in the medium term.

1. The 2013 updated joint Bank-Fund debt sustainability analysis (DSA) reconfirms the moderate risk rating found in the last DSA (country report No.1 2/329). The Kyrgyz Republic is classified as a large remittances case. The update takes into account revised macroeconomic assumptions and data on new borrowing[2]. Debt burden indicators under the baseline scenario with remittances show that Kyrgyz is at moderate risk. The macroeconomic assumptions are based on a revised growth path, following the sharp contraction in 2012, and a slower recovery of the current account deficit in the medium term. The lower gold price throughout the medium term and significantly higher imports related to public investment programs (PIPs) funded by foreign loans, grants, and FDI, are additional factors affecting the current account. In the long run, exports and imports are projected to grow faster, because of (i) the extension of the life of the Kumtor gold mine by 5 years to 2026 and new gold mines coming online, (ii) higher domestic demand resulting from the higher income from gold exports, (iii) growth of transportation services following the completion of large scale infrastructure projects (e.g. high ways).

Table 1. Kyrgyz Republic: Underlying Long-Term DSA Assumptions

	2012 DSA	2013 DSA Update
Real GDP growth (in percent)	5.0	5.0
Growth of exports of G&S (U.S. dollar terms, in percent)	6.4	8.6
Growth of imports of G&S (U.S. dollar terms, in percent)	6.5	7.3
Primary Deficit (percent of GDP)	1.7	1.7
Inflation rate	4.0	4.0

Sources: Kyrgyz authorities and IMF staff projections.

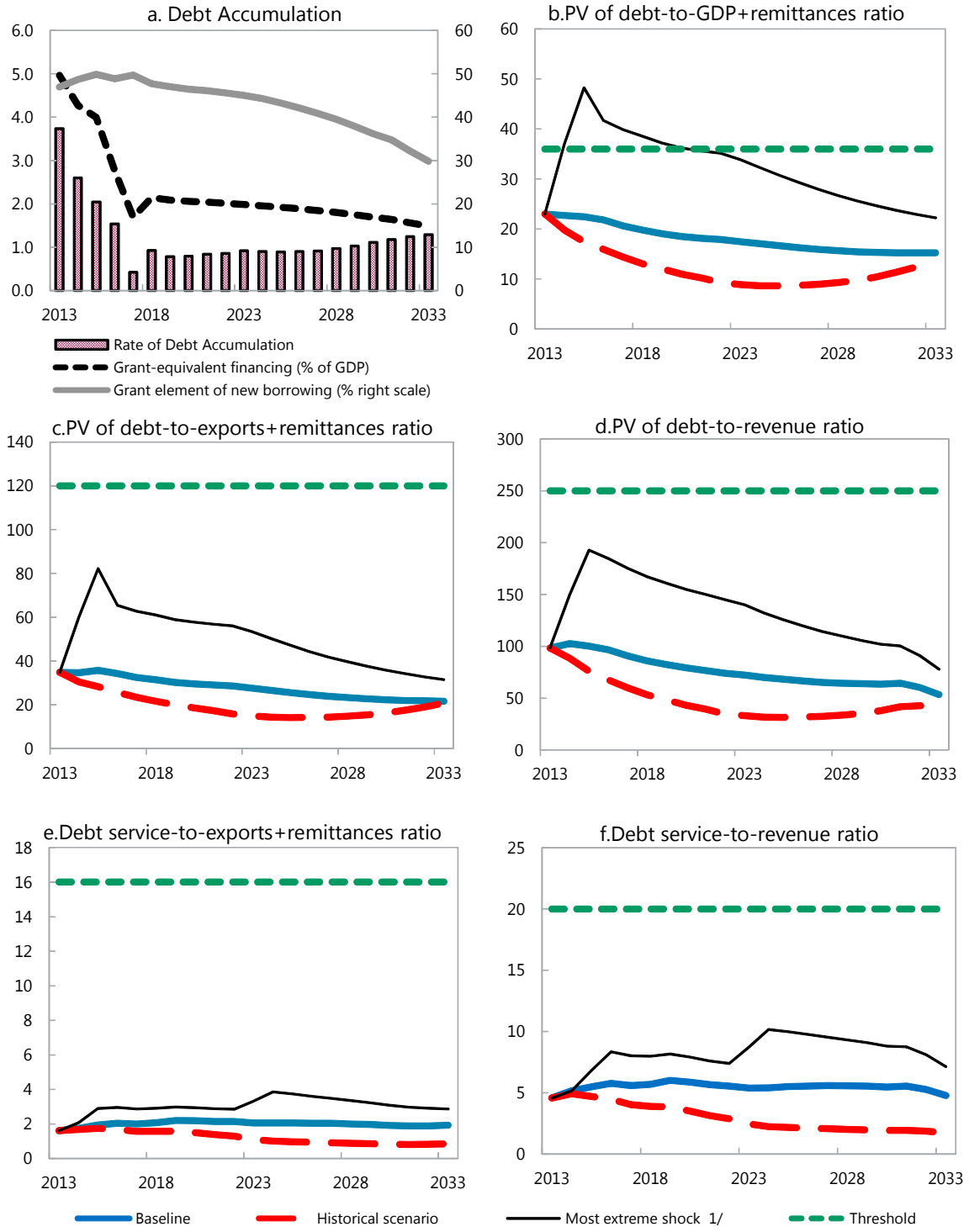
2. The debt outlook is sustainable but remains vulnerable to large external shocks, with remittance an important factor in mitigating risks of debt distress. All external debt burden indicators in the baseline scenario are below the thresholds. However, stress tests indicate that the Kyrgyz Republic remains vulnerable to large external shocks, especially in light of increased

[1] The average CPIA score 2010–12 is 3.62, while policy performance category is medium.

[2] Only part (US$200 million) of the Chinese PIP loans is included in the DSA. Assuming full disbursement of the US$800 million PIP loans would not change the DSA findings.

borrowing related to PIP projects over the medium-term. The present value (PV) of debt to GDP ratio in the most extreme shock (a shock to non-debt flows (FDI or net transfers in 2014 and 2015) breaches the threshold. Steady growth of remittances is essential for the Kyrgyz Republic to mitigate debt-related risks, including lower gold prices, changing global oil and food prices, a protracted slowdown in Russia and Kazakhstan, unfavorable developments in advanced and other emerging economies and the security situation in the region. The PV of external public debt to GDP ratio is projected to decline gradually from 32.7 percent in 2013 to 28.3 percent in 2033. Maintaining a prudent borrowing strategy remains key to keep the debt outlook sustainable, especially related to large projects such as the Kambarata hydro power plant.

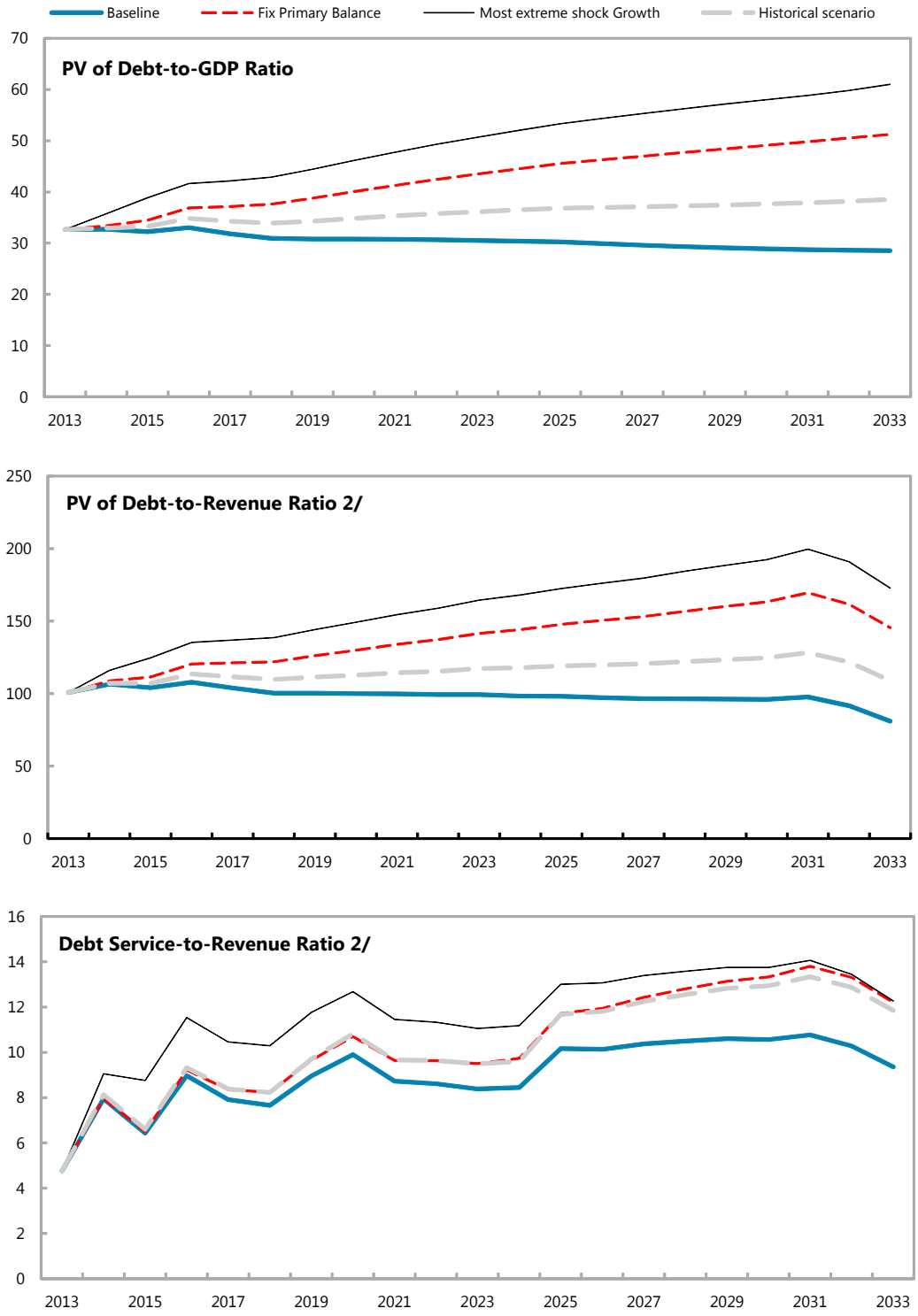
Figure 1. Kyrgyz Republic: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios Including Remittances, 2013–33



Sources: Kygyz authorities and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a Non-debt flows shock; in c. to a Non-debt flows shock; in d. to a Non-debt flows shock; in e. to a Non-debt flows shock and in figure f. to a Non-debt flows shock

Figure 2. Kyrgyz Republic: Indicators of Public Debt under Alternatives Scenarios Including Remittances, 2013–33 1/



Sources: Kyrgyz authorities and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023.
2/ Revenues are defined inclusive of grants.

Table 1. Kyrgyz Republic: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010–33

	Actual					Estimate	Projections								
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	59.7	49.4	50.0			48.0	48.3	48.1	48.7	46.4	44.9		42.9	39.0	
of which: foreign-currency denominated	56.8	46.2	47.9			45.3	45.1	45.0	44.3	41.5	39.7		34.1	29.2	
Change in public sector debt	1.7	-10.3	0.6			-2.0	0.3	-0.2	0.7	-2.3	-1.5		-0.4	-0.2	
Identified debt-creating flows	4.5	-9.9	3.2			0.0	-0.4	-1.2	-1.3	-1.1	-1.3		-0.5	-0.3	
Primary deficit	5.2	3.7	4.4	2.3	2.0	4.3	3.2	1.7	1.5	1.6	1.7	2.3	1.7	1.7	1.7
Revenue and grants	30.5	31.8	34.5			32.4	30.7	31.0	30.6	30.7	30.8		30.7	34.5	
of which: grants	2.8	3.0	2.6			2.0	1.9	1.9	1.0	1.0	0.9		0.8	0.6	
Primary (noninterest) expenditure	35.7	35.5	38.9			36.7	33.9	32.6	32.1	32.3	32.6		32.4	36.2	
Automatic debt dynamics	-0.6	-13.5	-1.1			-4.3	-3.6	-2.9	-2.6	-2.7	-2.9		-2.2	-1.9	
Contribution from interest rate/growth differential	0.7	-3.3	0.5			-3.7	-3.0	-2.5	-2.4	-2.5	-2.4		-2.2	-1.4	
of which: contribution from average real interest rate	0.4	0.1	0.1			0.0	0.0	0.0	-0.2	-0.2	-0.2		-0.1	0.5	
of which: contribution from real GDP growth	0.3	-3.4	0.4			-3.6	-2.9	-2.4	-2.3	-2.3	-2.2		-2.1	-1.9	
Contribution from real exchange rate depreciation	-1.3	-10.2	-1.7			-0.6	-0.6	-0.4	-0.1	-0.3	-0.5		...	...	
Other identified debt-creating flows	-0.1	-0.1	-0.1			0.0	0.0	0.0	-0.2	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.1	-0.1	-0.1			0.0	0.0	0.0	-0.2	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-2.8	-0.4	-2.6			-2.0	0.7	1.0	1.9	-1.2	-0.3		0.2	0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	32.4			32.7	32.7	32.2	33.0	31.8	31.0		30.4	28.3	
of which: foreign-currency denominated	...	...	30.3			29.9	29.6	29.2	28.6	27.0	25.7		21.6	18.5	
of which: external	...	...	29.6			29.9	29.6	29.2	28.6	27.0	25.7		21.6	18.5	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	8.7	7.1	11.3			6.9	7.0	5.0	5.6	6.0	6.2		7.6	5.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	93.8			100.9	106.6	104.1	107.8	103.8	100.4		98.9	82.1	
PV of public sector debt-to-revenue ratio (in percent)	...	...	101.4			107.5	113.6	110.8	111.5	107.2	103.5		101.6	83.6	
of which: external 3/	...	...	92.8			98.5	102.6	100.3	96.7	90.8	86.0		72.3	54.5	
Debt service-to-revenue and grants ratio (in percent) 4/	7.5	7.0	15.5			4.8	8.0	6.4	9.0	7.9	7.7		8.3	9.4	
Debt service-to-revenue ratio (in percent) 4/	8.2	7.8	16.8			5.1	8.5	6.8	9.3	8.2	7.9		8.5	9.6	
Primary deficit that stabilizes the debt-to-GDP ratio	3.5	14.0	3.8			6.3	2.9	1.9	0.8	4.0	3.2		2.1	1.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-0.5	6.0	-0.9	4.1	3.6	7.8	6.5	5.3	5.0	5.0	5.0	5.8	5.0	5.0	5.0
Average nominal interest rate on forex debt (in percent)	1.1	1.2	1.2	1.2	0.3	1.2	1.5	1.6	1.4	1.4	1.4	1.4	1.5	1.8	1.6
Average real interest rate on domestic debt (in percent)	2.5	-7.3	7.6	-0.1	6.2	13.4	7.1	3.9	2.5	1.9	0.0	4.8	0.0	0.0	0.0
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.3	-18.9	-3.6	-7.2	7.9	-1.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	10.0	22.4	7.4	10.8	7.1	6.5	6.5	6.0	5.4	5.7	5.4	5.9	4.0	4.0	4.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.2	0.1
Grant element of new external borrowing (in percent)	...	...	...	...	...	47.0	48.7	49.9	48.9	49.7	47.7	48.6	45.0	29.8	...

Sources: Country authorities; and staff estimates and projections.
1/ Covers general government and gross debt is used.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public Debt, 2013–33

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	33	33	32	33	32	31	30	28
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	33	33	33	35	34	34	36	38
A2. Primary balance is unchanged from 2013	33	33	34	37	37	38	43	51
A3. Permanently lower GDP growth 1/	33	33	33	35	34	34	40	59
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	33	36	39	42	42	43	51	61
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	33	33	34	35	34	33	32	29
B3. Combination of B1-B2 using one half standard deviation shocks	33	34	36	38	38	38	43	49
B4. One-time 30 percent real depreciation in 2014	33	44	42	42	40	39	35	31
B5. 10 percent of GDP increase in other debt-creating flows in 2014	33	39	38	39	37	36	35	31
PV of Debt-to-Revenue Ratio 2/								
Baseline	101	107	104	108	104	100	99	80
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	101	107	107	114	112	110	117	108
A2. Primary balance is unchanged from 2013	101	109	111	120	121	122	141	145
A3. Permanently lower GDP growth 1/	101	108	107	113	111	110	129	167
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	101	116	125	135	137	139	164	172
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	101	109	111	115	111	107	105	83
B3. Combination of B1-B2 using one half standard deviation shocks	101	111	114	123	122	122	139	139
B4. One-time 30 percent real depreciation in 2014	101	145	137	138	132	126	115	88
B5. 10 percent of GDP increase in other debt-creating flows in 2014	101	126	123	126	121	117	113	88
Debt Service-to-Revenue Ratio 2/								
Baseline	5	8	6	9	8	8	8	9
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	5	8	7	9	8	8	9	12
A2. Primary balance is unchanged from 2013	5	8	7	9	8	8	9	12
A3. Permanently lower GDP growth 1/	5	8	7	9	8	8	10	13
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	5	8	7	10	9	9	11	14
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	5	8	7	9	8	8	8	10
B3. Combination of B1-B2 using one half standard deviation shocks	5	8	7	10	9	9	10	12
B4. One-time 30 percent real depreciation in 2014	5	9	9	12	10	10	11	12
B5. 10 percent of GDP increase in other debt-creating flows in 2014	5	8	7	10	8	8	9	10

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table 3a. Kyrgyz Republic: External Debt Sustainability Framework, Baseline Scenario, 2010–33

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**92.7**	**76.7**	**81.3**			**74.4**	**72.6**	**71.3**	**69.7**	**66.0**	**63.0**		**56.8**	**66.9**	
of which: public and publicly guaranteed (PPG)	55.9	45.6	47.2			45.3	45.1	45.0	44.3	41.5	39.7		34.1	29.2	
Change in external debt	3.5	-16.1	4.6			-6.9	-1.8	-1.3	-1.6	-3.7	-3.1		-0.2	2.1	
Identified net debt-creating flows	-4.8	-25.7	6.3			-2.6	-1.0	-1.1	-3.2	-4.3	-3.5		-2.8	-3.6	
Non-interest current account deficit	**5.6**	**6.0**	**14.6**	**3.7**	**7.3**	**8.7**	**13.8**	**13.1**	**10.2**	**7.7**	**3.5**		**2.4**	**-1.5**	1.3
Deficit in balance of goods and services	29.0	27.6	44.9			36.4	40.1	38.4	36.4	33.9	29.1		24.5	18.7	
Exports	52.7	54.7	54.7			55.7	55.2	51.9	52.4	52.5	52.2		54.8	64.5	
Imports	81.7	82.3	99.6			92.2	95.3	90.3	88.9	86.4	81.3		79.2	83.1	
Net current transfers (negative = inflow)	-29.0	-29.6	-31.8	-24.0	7.6	-30.3	-30.2	-29.8	-31.1	-30.6	-29.9		-23.7	-21.4	-23.6
of which: official	-1.6	-1.3	-1.0			0.0	0.0	0.0	0.0	0.0	-0.2		-0.1	-0.1	
Other current account flows (negative = net inflow)	5.6	8.1	1.5			2.5	3.9	4.5	4.9	4.4	4.3		1.7	1.3	
Net FDI (negative = inflow)	**-9.1**	**-11.2**	**-5.7**	**-6.0**	**2.9**	**-7.4**	**-12.2**	**-12.7**	**-12.1**	**-10.8**	**-5.9**		**-4.9**	**-3.7**	-4.5
Endogenous debt dynamics 2/	**-1.3**	**-20.5**	**-2.5**			**-3.9**	**-2.5**	**-1.5**	**-1.2**	**-1.2**	**-1.1**		**-0.3**	**1.5**	
Contribution from nominal interest rate	0.8	0.5	0.7			1.7	1.9	2.1	2.1	2.0	2.0		2.3	4.6	
Contribution from real GDP growth	0.4	-4.3	0.7			-5.7	-4.4	-3.6	-3.3	-3.2	-3.0		-2.7	-3.1	
Contribution from price and exchange rate changes	-2.5	-16.7	-3.9			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**8.3**	**9.6**	**-1.7**			**-4.2**	**-0.8**	**-0.2**	**1.6**	**0.6**	**0.5**		**2.6**	**5.7**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	63.7			59.1	57.1	55.5	54.0	51.5	49.0		44.4	56.1	
In percent of exports	...	...	116.4			106.0	103.4	107.0	103.1	98.1	93.9		81.0	87.1	
PV of PPG external debt	**...**	**...**	**29.6**			**29.9**	**29.6**	**29.2**	**28.6**	**27.0**	**25.7**		**21.6**	**18.5**	
In percent of exports	**...**	**...**	**54.2**			**53.7**	**53.5**	**56.2**	**54.6**	**51.4**	**49.2**		**39.5**	**28.7**	
In percent of government revenues	**...**	**...**	**92.8**			**98.5**	**102.6**	**100.3**	**96.7**	**90.8**	**86.0**		**72.3**	**53.4**	
Debt service-to-exports ratio (in percent)	**19.1**	**8.4**	**15.9**			**17.0**	**14.0**	**15.7**	**16.8**	**15.7**	**15.5**		**17.7**	**30.5**	
PPG debt service-to-exports ratio (in percent)	**3.6**	**2.9**	**9.7**			**2.5**	**2.7**	**3.1**	**3.3**	**3.2**	**3.3**		**2.9**	**2.6**	
PPG debt service-to-revenue ratio (in percent)	**6.9**	**5.5**	**16.7**			**4.6**	**5.2**	**5.5**	**5.8**	**5.6**	**5.7**		**5.4**	**4.8**	
Total gross financing need (Billions of U.S. dollars)	0.3	0.0	1.1			0.8	0.7	0.7	0.6	0.5	0.6		1.1	4.4	
Non-interest current account deficit that stabilizes debt ratio	2.2	22.1	10.0			15.5	15.6	14.4	11.8	11.4	6.5		2.6	-3.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	-0.5	6.0	-0.9	4.1	3.6	7.8	6.5	5.3	5.0	5.0	5.0	5.8	5.0	5.0	5.0
GDP deflator in US dollar terms (change in percent)	2.8	22.0	5.4	11.0	11.0	4.1	3.4	2.9	2.3	2.6	2.9	3.0	2.0	2.0	2.0
Effective interest rate (percent) 5/	0.9	0.7	1.0	1.2	0.4	2.4	2.8	3.1	3.1	3.2	3.2	3.0	4.4	7.6	5.3
Growth of exports of G&S (US dollar terms, in percent)	-1.1	34.1	4.5	20.0	18.4	14.4	9.2	1.9	8.5	7.9	7.4	8.2	9.2	8.5	8.6
Growth of imports of G&S (US dollar terms, in percent)	6.0	30.2	26.4	26.1	23.2	3.8	13.9	2.7	5.7	4.8	1.6	5.4	7.8	7.3	7.3
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	47.0	48.7	49.9	48.9	49.7	47.7	48.6	45.0	29.8	40.7
Government revenues (excluding grants, in percent of GDP)	27.7	28.8	31.9			30.4	28.8	29.1	29.6	29.7	29.9		29.9	34.6	30.2
Aid flows (in Billions of US dollars) 7/	0.3	0.4	0.3			0.6	0.5	0.5	0.4	0.2	0.4		0.5	0.7	
of which: Grants	0.1	0.2	0.2			0.1	0.2	0.2	0.1	0.1	0.1		0.1	0.2	
of which: Concessional loans	0.2	0.2	0.1			0.4	0.4	0.4	0.3	0.1	0.3		0.3	0.5	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.0	4.3	4.0	2.8	1.7	2.1		2.0	1.5	1.8
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			59.6	63.1	65.2	60.5	69.7	61.8		58.0	42.6	53.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	4.8	6.2	6.5			7.3	8.0	8.7	9.3	10.0	10.8		15.4	30.4	
Nominal dollar GDP growth	2.4	29.3	4.4			12.2	10.2	8.4	7.4	7.7	8.0	9.0	7.1	7.1	7.1
PV of PPG external debt (in Billions of US dollars)			1.9			2.1	2.3	2.5	2.6	2.7	2.8		3.3	5.6	
(PVt-PVt-1)/GDPt-1 (in percent)						3.7	2.6	2.0	1.5	0.4	0.9	1.9	0.9	1.3	1.0
Gross workers' remittances (Billions of US dollars)	1.3	1.8	2.0			2.2	2.4	2.6	2.9	3.1	3.2		3.6	6.5	
PV of PPG external debt (in percent of GDP + remittances)	...	...	22.6			23.0	22.7	22.5	21.8	20.6	19.8		17.5	15.2	
PV of PPG external debt (in percent of exports + remittances)	...	...	34.6			34.8	34.6	35.7	34.3	32.4	31.4		27.6	21.5	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	6.2			1.6	1.7	2.0	2.0	2.0	2.1		2.1	1.9	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3b. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt Including Remittances, 2013–33

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to-GDP+remittances ratio								
Baseline	23	23	22	22	21	20	**17**	15
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	23	20	17	16	15	13	**9**	14
A2. New public sector loans on less favorable terms in 2013-2033 2	23	23	24	24	23	23	**23**	26
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	23	23	24	23	22	21	**19**	16
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	23	24	26	25	24	23	**20**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	23	23	23	23	21	21	**18**	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	23	37	48	42	40	39	**34**	22
B5. Combination of B1-B4 using one-half standard deviation shocks	23	34	40	35	34	32	**29**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	23	29	29	28	26	25	**23**	20
PV of debt-to-exports+remittances ratio								
Baseline	35	35	36	34	32	31	**28**	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	35	30	28	26	24	22	**15**	21
A2. New public sector loans on less favorable terms in 2013-2033 2	35	36	38	38	37	36	**37**	37
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	35	34	35	34	32	31	**27**	21
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	35	39	43	41	39	38	**33**	24
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	35	34	35	34	32	31	**27**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	35	60	82	65	63	61	**53**	31
B5. Combination of B1-B4 using one-half standard deviation shocks	35	52	60	51	49	47	**41**	26
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	35	34	35	34	32	31	**27**	21
PV of debt-to-revenue ratio								
Baseline	98	103	100	97	91	86	**72**	53
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	98	89	76	68	60	54	**34**	42
A2. New public sector loans on less favorable terms in 2013-2033 2	98	106	108	107	103	100	**97**	91
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	98	107	110	106	99	95	**79**	58
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	98	109	114	110	103	98	**83**	57
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	98	105	105	101	95	90	**76**	56
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	98	150	193	185	175	167	**140**	78
B5. Combination of B1-B4 using one-half standard deviation shocks	98	141	165	158	150	143	**119**	71
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	98	142	139	134	126	120	**100**	74

Table 3b. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt Including Remittances, 2013–33 (concluded)

Debt service-to-exports+remittances ratio								
Baseline	2	2	2	2	2	2	**2**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	2	2	2	2	2	2	**1**	1
A2. New public sector loans on less favorable terms in 2013-2033 2	2	2	2	2	2	2	**2**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	2	2	2	2	2	2	**2**	2
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	2	2	2	2	2	2	**2**	2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	2	2	2	2	2	2	**2**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	2	2	3	3	3	3	**3**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	2	2	2	2	**3**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	2	2	2	2	2	2	**2**	2
Debt service-to-revenue ratio								
Baseline	5	5	5	6	6	6	**5**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	5	5	5	4	4	4	**2**	2
A2. New public sector loans on less favorable terms in 2013-2033 2	5	5	5	6	6	6	**6**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	5	5	6	6	6	6	**6**	5
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	5	5	6	6	6	6	**6**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	5	5	6	6	6	6	**6**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	5	5	7	8	8	8	**9**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	5	5	7	8	7	7	**8**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	5	7	8	8	8	8	**8**	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	39	39	39	39	39	39	**39**	39

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3c. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	30	30	29	29	27	26	**22**	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	30	25	22	20	18	16	**10**	15
A2. New public sector loans on less favorable terms in 2013-2033 2	30	31	31	32	30	30	**29**	31
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	30	31	32	31	30	28	**24**	20
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	30	31	33	33	31	29	**25**	20
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	30	30	31	30	28	27	**23**	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	30	43	56	55	52	50	**42**	27
B5. Combination of B1-B4 using one-half standard deviation shocks	30	41	48	47	44	43	**36**	25
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	30	41	40	40	37	36	**30**	26
PV of debt-to-exports ratio								
Baseline	54	54	56	55	51	49	**39**	29
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	54	46	43	38	34	31	**19**	23
A2. New public sector loans on less favorable terms in 2013-2033 2	54	55	61	61	58	57	**53**	49
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	54	53	55	54	51	49	**39**	28
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	54	61	69	67	63	61	**49**	33
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	54	53	55	54	51	49	**39**	28
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	54	78	108	104	99	96	**77**	42
B5. Combination of B1-B4 using one-half standard deviation shocks	54	71	82	79	75	72	**58**	34
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	54	53	55	54	51	49	**39**	28
PV of debt-to-revenue ratio								
Baseline	98	103	100	97	91	86	**72**	53
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	98	89	76	68	60	54	**34**	42
A2. New public sector loans on less favorable terms in 2013-2033 2	98	106	108	107	103	100	**97**	91
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	98	107	110	106	99	95	**79**	58
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	98	109	114	110	103	98	**83**	57
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	98	105	105	101	95	90	**76**	56
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	98	150	193	185	175	167	**140**	78
B5. Combination of B1-B4 using one-half standard deviation shocks	98	141	165	158	150	143	**119**	71
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	98	142	139	134	126	120	**100**	74

Table 3c. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33 (concluded)

(In percent)

Debt service-to-exports ratio								
Baseline	2	3	3	3	3	3	**3**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	2	3	3	3	2	2	**1**	1
A2. New public sector loans on less favorable terms in 2013-2033 2	2	3	3	3	3	4	**3**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	2	3	3	3	3	3	**3**	3
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	2	3	3	4	4	4	**3**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	2	3	3	3	3	3	**3**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	2	3	4	5	5	5	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	2	3	3	4	4	4	**4**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	2	3	3	3	3	3	**3**	3
Debt service-to-revenue ratio								
Baseline	5	5	5	6	6	6	**5**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	5	5	5	4	4	4	**2**	2
A2. New public sector loans on less favorable terms in 2013-2033 2	5	5	5	6	6	6	**6**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	5	5	6	6	6	6	**6**	5
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	5	5	6	6	6	6	**6**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	5	5	6	6	6	6	**6**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	5	5	7	8	8	8	**9**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	5	5	7	8	7	7	**8**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	5	7	8	8	8	8	**8**	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	39	39	39	39	39	39	**39**	39

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.